

Fractional Structure and Möbius Transformation — Part I: Double Inversion in Division and the Phase of Twist

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Abstract

We formalize “double inversion” in ordinary division as a joint action that (i) inverts the *operator* (division $\rightarrow$ multiplication) and (ii) inverts the *operand* (the divisor $y \mapsto y^{-1}$). Writing integers as ratios $a = a/1$ foregrounds the peculiar role of 1: it is the multiplicative unit (anchor), the unique positive fixed point of the reciprocal map $I(x) = 1/x$ with $I'(1) = -1$, and the mirror center in logarithmic coordinates ($\log x \mapsto -\log x$). We model division-by- y as the involutive transform $T : (\div, y) \mapsto (\times, y^{-1})$ and situate it inside the group of linear fractional (Möbius) maps on the projective line $\mathbb{P}^1$. Geometrically, I exchanges 0 and ∞ while fixing 1, realizing a “half-twist” analogous to the Möbius band. This Part I develops the minimal algebraic-geometric framework and a phase interpretation of the twist; Part II argues that the symmetry’s critical locus is not at 0 but at 1.

Keywords Möbius transformation; linear fractional map; double inversion; projective line; reciprocal symmetry; logarithmic coordinate; twist phase.

1 Introduction

Division is routinely rewritten as multiplication by the reciprocal:

$$a \div y = a \times y^{-1}.$$

We call the joint switch of the operator (“ $\div \rightarrow \times$ ”) and the operand (“ $y \rightarrow y^{-1}$ ”) a *double inversion*. Expressing integers as ratios ($a = a/1$) reveals the special status of 1, which will act as a pivot between 0 and ∞ .

2 Algebraic formalization

Define the transform

$$T : (\div, y) \longmapsto (\times, y^{-1}).$$

Then T is an involution ($T^2 = \text{id}$) and

$$\frac{a/b}{c/d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}.$$

The map $I(x) = 1/x$ has, on $\mathbb{R}_{>0}$, the unique fixed point $x = 1$ with derivative $I'(1) = -1$, indicating orientation reversal without expansion/contraction. In log-coordinate $s = \log x$, reciprocity becomes the reflection $s \mapsto -s$, so $x = 1$ (i.e., $s = 0$) is the mirror center.

3 Möbius viewpoint on ratios

Ratios correspond to points of the projective line $\mathbb{P}^1$. Linear fractional (Möbius) maps $z \mapsto \frac{az+b}{cz+d}$ act via matrices ab
 cd (up to scale). The reciprocal $I(z) = 1/z$ is realized by $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, which exchanges 0 and ∞ and fixes 1. Thus the algebraic double inversion embeds naturally into a projective symmetry that implements a “half-twist” in orientation.

4 Twist phase (phase-of-twist interpretation)

Under $s = \log x$, the reciprocal symmetry is $s \mapsto -s$. We interpret $x = 1$ (the mirror $s = 0$) as the *phase center* of the twist: numbers $x > 1$ and $x < 1$ are exchanged as reciprocal duals, with 1 as the hinge. In this sense, 1 plays a role different from generic integers: it is the pivot where operator inversion and operand inversion couple.

5 Math–philosophy correspondence (minimal schema)

- **Operator inversion** ($\div \rightarrow \times$) $\leftrightarrow$ switch of *action* (how we relate magnitudes).
- **Operand inversion** ($y \rightarrow y^{-1}$) $\leftrightarrow$ switch of *perspective* (inside/outside of scale).

- **Pivot 1** (fixed point with $I'(1) = -1$) $\leftrightarrow$ *hinge of duality* (the mirror center in information/scale space).

6 Conclusion & Outlook to Part II

We captured division as a double inversion and located it within Möbius symmetry on $\mathbb{P}^1$, highlighting the special role of 1. Part II advances the *critical-one hypothesis*: the effective singular locus of reciprocal symmetry sits at 1 (the “egg of infinity”), not at 0.